



HYPERSDR V1.0

AssemblyAI Voice Agent Hackathon • Lablab.ai

● [Live on Render](#)

September 30, 2026

🏆 Official Hackathon Submission • AssemblyAI Voice Agent API Track

HyperSDR

Autonomous Voice Sales SDR Built on **AssemblyAI**
Voice Agent API for 212 Luxury Boutique Hotels &
Historic Inns.

VOICE & SPEECH STACK

Universal-3.5 Pro

George Voice • Sub-second VAD

FUNCTION CALLING

Google Calendar API

Real Meet Links & 2-Way Sheet DB

TARGET MARKET IMPACT

212 US Innkeepers

\$338,000/yr Commission Leakage

THE CORE CHALLENGE

The High-Stakes Vulnerability of Boutique Hotels

18% – 22%

Bleeding Revenue to OTA Commissions

Independent boutique hotels and historic inns lose hundreds of thousands of dollars annually to Expedia and Booking.com—even when **repeat guests** who already know the inn rebook through third-party travel platforms.

🔥 Average 20-room inn loses **\$216k – \$338k / year** in avoidable OTA commission cuts.

Google 3-Pack

Fragile Local Search Reputation

A single unaddressed guest complaint posted publicly on Google Maps can drop a boutique inn from the coveted Top 3 Local Pack, instantly wiping out organic weekend guest direct bookings and wedding inquiries.

⚠️ Overburdened innkeepers lack automated private post-stay triage systems before feedback hits Google.

AUTONOMOUS VOICE SOLUTION

Enter HyperSDR: Voice SDR for Innkeepers

01

Tailored Review Hooks

Alex (SDR) opens each call with verified guest intelligence: Chef Jim's culinary breakfast, private whirlpools, or red rock labyrinths to establish instant rapport with owners.

🌟 Zero generic spam pitches

02

Live Tool Booking

When the innkeeper agrees, Alex invokes Google Calendar API directly in-call, creating real calendar events and generating authentic Google Meet video conference links.

📅 In-flight Google Calendar sync

03

Speech Intelligence

Upon call termination, HyperSDR executes automated BANT scoring, extracts key takeaways, and performs two-way live write-backs to Google Sheets & Excel.

📊 Zero post-call data entry

TECHNICAL ARCHITECTURE

Built on AssemblyAI Edge Voice Infrastructure



TARGET MARKET INTELLIGENCE

212 Pre-Enriched US Boutique Hotel Leads

Deep Dossier Enrichment Schema

- ✓ **Decision Maker:** Named Proprietor, Owner & Innkeeper
- ✓ **Contact Points:** Verified Phone & Direct Email
- ✓ **Reputation:** Google Rating & Review Count (3-Pack status)
- ✓ **Cold Hook:** Signature Amenity & Verbatim Guest Quote
- ✓ **Financial ROI:** Calculated Annual OTA Commission Leakage

212 Verified Leads

High-value destinations across Napa Valley, Sedona, Shenandoah, Maine, Florida & Vermont.

Eastern Timezone (EDT) 114 Properties

Pacific Timezone (PDT) 58 Properties

Central & Mountain (CDT/MDT) 40 Properties

⚡ **Smart Regional Clock Engine:** Automatically prioritizes properties located in active 9:00 AM - 5:00 PM local calling windows.

LIVE FUNCTION CALLING EXECUTION

Real Google Calendar & Google Meet Booking

TOOL 01

book_calendar_slot

Schedules 15-minute strategy syncs on Google Calendar API v3 with timezone offset calculation and instant attendee invites.

TOOL 02

calculate_ota_commission

Computes room count x ADR x occupancy x OTA commission rate to quantify exact recovered profit during the pitch.

TOOL 03

mark_call_disposition

Records qualification score (1-100), key takeaways, objection notes, and updates CRM database instantly.

✓ **Verified Production Booking (Google Calendar API v3):**

Event ID:
ol9ln8hgqu0ag79tad96cg470c

Google Meet URL:
<https://meet.google.com/eni-uzia-pyd>

Timezone & Slot:
Thu, Oct 1 @ 2:00 PM PDT

AUDIO ENGINEERING & UX

Comfort Noise Layer for Natural Voice Dynamics

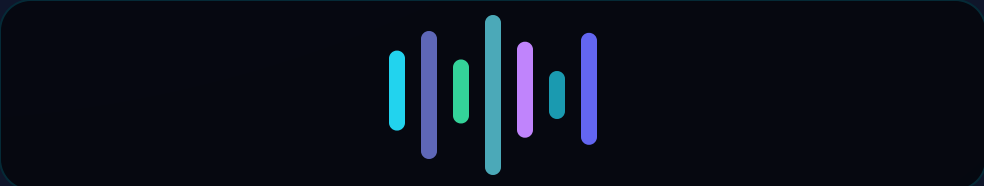
Eliminating Robotic Dead Air

Traditional AI phone bots feel synthetic and robotic due to pure digital silence during tool execution or network hops. HyperSDR introduces a LiveKit-style pink/brown noise WebAudio generator to give Alex subtle, acoustic warmth.

-  Subtle Office
-  Phone Line Warmth
-  Room Acoustics

Dual Canvas Audio Waveform Visualizers

Real-time browser microphone FFT analysis + AssemblyAI inbound agent playback monitoring.



Microphone Ingest (16kHz PCM16) • Speaker Output Waveform • Master Comfort Noise Volume Slider

POST-CALL ANALYTICS & CLOUD CRM

Post-Call Intelligence & Live Google Sheet DB

Post-Call BANT Speech Intelligence

💰 Budget (OTA Commission Leakage):	25 / 25
👑 Authority (Proprietor/Innkeeper):	25 / 25
🎯 Need (Google 3-Pack Defense):	25 / 25
📅 Timeline (Strategy Sync Locked):	25 / 25
Total Lead Qualification Score:	100 / 100 (HOT PROSPECT)

Google Sheets API v4 Real-Time Sync

Direct two-way spreadsheet database write-back without edge caching or manual refresh delays:

- **Row 2 Updated:** The Inn On First (USA-VIP-0001)
- **Call Status:** COMPLETED
- **Outcome:** meeting_booked (YES)
- **Meeting Slot:** Thu, Oct 1 @ 2:00 PM PDT
- **Meet Link:** <https://meet.google.com/eni-uzia-pyd>

OAuth 2.0 Refresh Token auto-renewing in cloud memory on Render.

BUSINESS CASE & MARKET OPPORTUNITY

Quantifiable ROI for Independent Hospitality

\$338k / yr

COMMISSION RECOVERY

Annual savings per 20-room inn by shifting repeat guests off Expedia & Booking.com into direct commission-free bookings.

Recovers 18-22% margin

0 Hours

STAFF OVERHEAD

Private 48-hour post-stay review routing and outreach pipeline runs automatically with zero hardware or software installs.

Autonomous front-desk support

100%

VOICE AI AUTOMATION

Sub-second AssemblyAI Universal-3.5 Pro STT handles qualifying, hook pitching, objection handling, and booking end-to-end.

Production-grade accuracy

 Ready for Hackathon Judges & Production Dialing

HyperSDR is Live & Deployed

Test the live web dialer, initiate real conversational phone calls with Alex, and witness instant Google Calendar & Google Sheet synchronization.

Launch Live App ↗

hypersdr-assemblyai.onrender.com

View GitHub Repo ↗

github.com/AbrarMuhtasim14/assemblyai-hypersdr

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